

practical no.9

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1 . What is a Data Science Pipeline? Explain its key stages with examples

A Data Science Pipeline is a sequence of steps used to collect, process, analyze, and build models from data to solve a problem.

Key Stages: Data Collection Gathering data from sources (databases, APIs, CSV files)

Example: Collecting customer data from an e-commerce website Data Cleaning Handling missing values, removing duplicates, fixing errors Example: Filling missing age values with mean

Exploratory Data Analysis (EDA) Understanding data using graphs and statistics

Example: Plotting sales distribution Feature Engineering Creating or modifying features for better performance Example: Extracting "day" from a date column

Model Building Training machine learning models Example: Using Linear Regression to predict prices

Model Evaluation Checking model accuracy and performance Example: Using accuracy, precision, recall

Deployment Making the model available for real-world use Example:

Recommendation system on a shopping app

2. What is Exploratory Data Analysis (EDA)?

EDA is the process of analyzing datasets to understand their structure, patterns, and relationships before building models.

Types of Analysis: Univariate Analysis (Single Variable) Focuses on one variable at a time

Helps understand distribution, central tendency, spread

Examples:

Histogram of age Mean, median, mode

👉 Helps answer:

Is data skewed? Are there outliers? Bivariate Analysis (Two Variables) Studies relationship between two variables

Examples:

Scatter plot (height vs weight) Correlation matrix

👉 Helps answer:

Are variables related? Positive or negative correlation?

3. Difference between Label Encoding and One-Hot Encoding

Feature	Label Encoding	One-Hot Encoding
Definition	Converts categories into numbers	Converts categories into binary columns
Example	Red=0, Blue=1, Green=2	Red=[1,0,0], Blue=[0,1,0]
Relationship	Implies order	No order implied
Columns	1 column	Multiple columns

When to Use: Label Encoding: When data is ordinal (e.g., Low, Medium, High) One-Hot Encoding: When data is nominal (e.g., colors, city names)

4. Why is Feature Scaling Required?

Feature scaling ensures all features are on the same scale so that no variable dominates others.

👉 Important for algorithms like:

KNN SVM Gradient Descent

Feature	Normalization	Standardization
Range	0 to 1	Mean = 0, Std = 1
Formula	Min-Max scaling	Z-score
Sensitive to outliers	Yes	Less
Use case	When data has fixed range	When data is normally distributed

5. What is Feature Engineering?

Feature Engineering is the process of creating new input features or modifying existing ones to improve model performance.

Why it is important: Makes patterns more visible Improves accuracy Helps model learn better

Examples: Date Feature Creation From "Date" extract Day, Month, Year Combining Features Area + Location Price prediction Encoding Categorical Data Convert text to numbers Log Transformation Reduce skewness in data How it improves performance: Better representation of data Reduces noise Captures hidden relationships

Example: Instead of using "Date", using "Weekend/Weekday" can improve prediction in sales models.

```
import pandas as pd
from sklearn.datasets import load_iris

# Load the Iris dataset
iris = load_iris()
```

<https://colab.research.google.com/#scrollTo=628f6c36&printMode=true>

```
df = pd.DataFrame(data=iris.data, columns=iris . feature names ,
                  target = iris . target ,
                  target_names = iris . target_names )
df[ ' target ' ] . apply(lambda x:
iris . target_names [ x ] )
print ("Iris dataset loaded successfully. " )
display(df. head() )
```

Iris dataset loaded successfully.

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	target_names
0	5.1	3.5	1.4	0.2	0	setosa
1	4.9	3.0	1.4	0.2	setosa	
2	4.7	3.2	1.3	0.2	setosa	
3	4.6	3.1	1.5	0.2	setosa	
4	5.0	3.6	1.4	0.2	setosa	

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